

FENRIR ARENA

Freelancer Project Requirements Document

Fenrir Project | Online Gaming Website

Category	Project Type	Total Files	Backend
Web Development	Browser Gaming Portal	17 source files	Node.js + SQLite

1. Project Overview

You are hired to build **Fenrir Arena**, a browser-based gaming portal for the Fenrir Project. Visitors can browse a catalog of casual games, play them directly in the browser, create an account, and compete on leaderboards. The site consists of 8 pages and 6 playable games, with a lightweight backend for accounts, scores, and leaderboards. All deliverables are source code files and documentation. No physical production is required.

Field	Value
Product name	Fenrir Arena
Tagline	Play. Compete. Conquer. (use verbatim)
Category	Web Development
Total pages	8
Total games	6
Backend framework	Node.js with Express
Database	SQLite

2. Brand Guidelines

2.1 Colour Palette

Use	Colour Name	Hex Code
Primary background	Charcoal Black	#1A1A1F
Surface / header	Wolf Gray	#2E2E38
Accent / buttons	Ember Orange	#FF7A30
Secondary accent	Frost Blue	#4FB6E8
Text	Off White	#F5F5F7

2.2 Mandatory Rules

- The site name "Fenrir Arena" must appear in the website header on every page
- The tagline "Play. Compete. Conquer." must appear verbatim on the Home Page and the About Page - do not rephrase
- The Home Page featured games section must show exactly 4 games
- The Game Library Page must show all 6 games with category filters: Arcade, Puzzle, Racing

- The layout must be usable at screen widths from 360px to 1920px

3. Pages (8 total)

Each page must be a separate file and must include the website header showing "Fenrir Arena". File format: HTML + CSS + JavaScript.

#	Page	File Name	Required Elements
1	Home	index.html	Hero banner, tagline, 4 featured games, Browse All Games button
2	Game Library	library.html	Grid of 6 games, filters (Arcade/Puzzle/Racing), game cards
3	Individual Game	game.html	Canvas, title, description, How to Play, leaderboard (top 10)
4	Login	login.html	Email field, password field, Login button, Sign Up link
5	Sign Up	signup.html	Username, email, password fields, Create Account button, Login link
6	User Profile	profile.html	Username display, best score table per game
7	About	about.html	Description text, tagline verbatim
8	Contact	contact.html	Name, email, message fields, Send Message button

4. Games (6 total)

Each game must run inside an HTML5 Canvas element, must be controllable using keyboard or mouse input as specified below, and must display the player's current score during play. Each game is embedded on its own Individual Game Page.

Game	Category	Controls
Brick Breaker	Arcade	Left/Right arrow keys move the paddle
Snake Classic	Arcade	Arrow keys change direction
Memory Match	Puzzle	Mouse click flips cards
2048 Tile Merge	Puzzle	Arrow keys merge tiles
Endless Runner	Racing	Spacebar triggers jump
Sky Dash	Racing	Arrow keys steer, spacebar boosts

Score Submission: when a game ends, the final score must be sent to the backend save score endpoint. The score is saved only if the player is logged in.

5. Backend (Node.js + SQLite)

5.1 Endpoints

Endpoint	Function
Sign up	Accepts username, email, password; stores password as a hashed value
Login	Accepts email and password; returns a session that persists while the tab is open
Save score	Accepts username, game name, score value, and date; stores the record
Leaderboard	Accepts a game name; returns the top 10 scores sorted highest to lowest

5.2 Data Fields

Record	Fields
User account	username, email, hashed password
Score record	username, game name, score value, date

6. File Naming Convention

Page files: [pagename].html / .css / .js (e.g. index.html, library.html, game.html, login.html, signup.html, profile.html, about.html, contact.html). Game files: games/[gamenam].js (e.g. games/brick-breaker.js, games/snake-classic.js, games/memory-match.js, games/2048-tile-merge.js, games/endless-runner.js, games/sky-dash.js).

7. Folder Structure

fenrir-arena/

index.html, library.html, game.html, login.html, signup.html, profile.html, about.html, contact.html

css/styles.css

games/ (brick-breaker.js, snake-classic.js, memory-match.js, 2048-tile-merge.js, endless-runner.js, sky-dash.js)

backend/server.js, backend/routes/accounts.js, backend/routes/scores.js, backend/database.sqlite

README.md

8. Responsive Design Requirements

The layout must be usable on screen widths from 360px (mobile) to 1920px (desktop). All 8 pages must apply the colour palette in Section 2.1. All pages must include the website header showing "Fenrir Arena".

9. Final Deliverables Checklist

Category	Description	Files
Pages	8 pages: Home, Library, Game, Login, Sign Up, Profile, About, Contact	8
Games	6 games: Brick Breaker, Snake Classic, Memory Match, 2048 Tile Merge, Endless Runner, Sky Dash	6
Backend	Accounts route + Scores route	2
Docs	README.md with local run instructions	1

Total source files: 17 (8 pages + 6 games + 2 backend routes + 1 README)